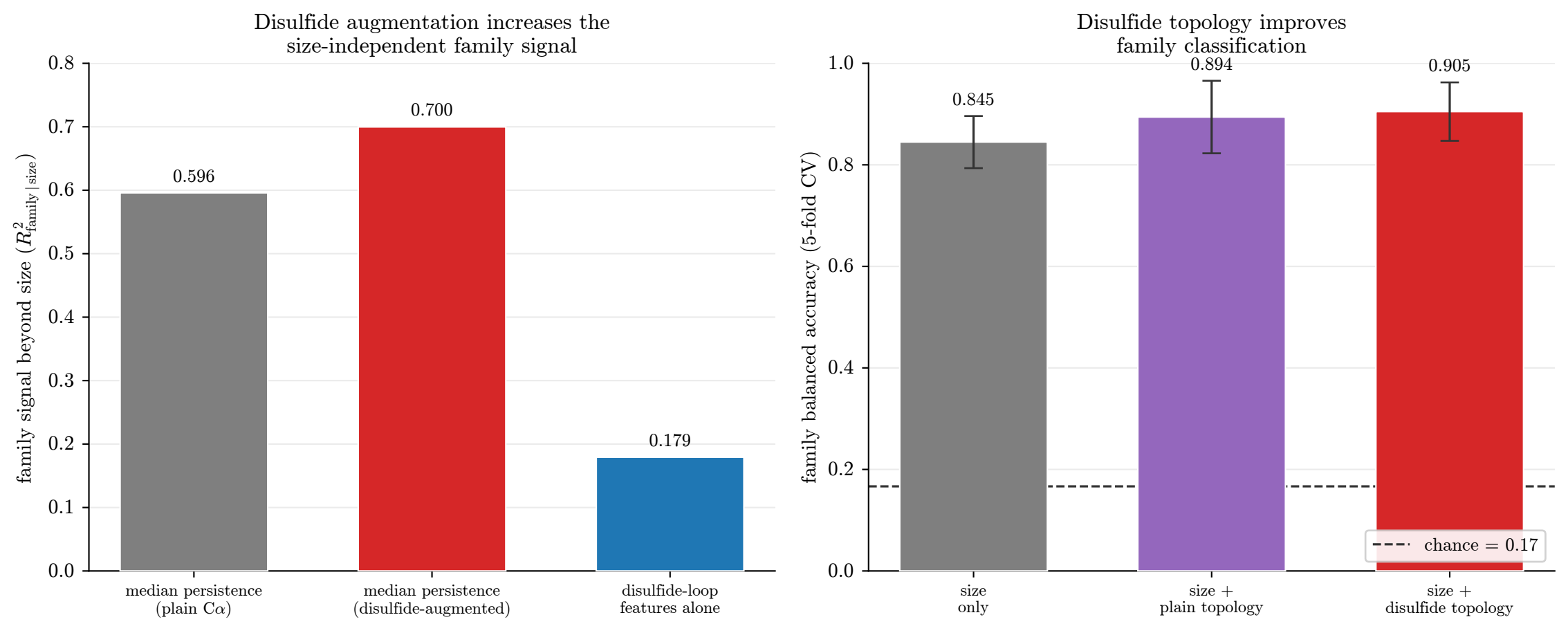
Disulfide augmentation increases the size-independent family signal
family signal beyond size ($R^2_{\text{family} \mid \text{size}}$)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.596
0.700
0.179
median persistence (plain Cα)
median persistence (disulfide-augmented)
disulfide-loop features alone
Disulfide topology improves family classification
family balanced accuracy (5-fold CV)
0.0
0.2
0.4
0.6
0.8
1.0
0.845
0.894
0.905
chance = 0.17
size only
size + plain topology
size + disulfide topology